

Chapter 11 - Audio Architecture Overview

Where the picture side has six chips feeding one compositor, the audio side has ten sound engines and a sample player feeding one **mixer**. The mixer always runs at 44,100 samples per second and sums the contributions of every active engine into a single stereo output. The chips themselves are independent - turning one on does not affect any other - and they can play together.

This chapter sketches the architecture and lists what lives where. Chapters 12–21 cover the individual engines; Chapter 22 shows how to drive them all from BASIC and from each CPU.

11.1 The sound engines

Chip / Engine	Chapter	Heritage
SoundChip + SFX	12	IE-native 10-channel synth + 4 sample channels
PSG / AY-3-8910	13	Sinclair Spectrum / Amstrad CPC / Atari ST
SN76489	14	TI / Sega Master System / BBC Micro
SID family	15	Commodore 64 (three SID chips: original 6581, 8580, plus IE's "SID+" extensions)
TED audio	16	Commodore 16 / Plus-4
POKEY + SAP	17	Atari 8-bit
AHX	18	Abyss High-Quality Music Format (Amiga)
MOD	19	Amiga ProTracker, four channels
WAV / sample	20	Generic 8/16-bit sample player
Paula DMA	21	Amiga Paula-style four-channel DMA sample player

Every engine has its own MMIO block. The SFX channels share a small block separate from the SoundChip's tone channels.

11.2 The mixer

All engines write into the same output stream. The mixer:

- Sums each engine's stereo contribution after applying that engine's own per-channel volume.
- Applies a global filter (low-/high-/band-pass) selected through the FILTER_* registers at 0xF0820–0xF0830.
- Applies optional overdrive (one OVERDRIVE_CTRL register at 0xF0A40) and reverb (REVERB_MIX at 0xF0A50 and REVERB_DECAY at 0xF0A54).
- Sends the result to the audio output of the Intuition Engine at 44,100 samples per second.

The master enable register is AUDIO_CTRL at 0xF0800. Bit 0 enables audio output; bit 1 freezes the mixer (useful while swapping presets).

11.3 The SoundChip channel block

The IE-native SoundChip exposes its channels through a uniform **flexible-channel** layout. Each channel is 0x40 bytes wide. Inside one channel:

Offset	Field	Meaning
0x00	FREQ	Frequency in 16.8 fixed-point Hz (value / 256.0 = Hz).
0x04	VOL	Volume, 0–255.
0x08	CTRL	Gate / control.
0x0C	DUTY	PWM duty cycle.
0x10	SWEEP	Pitch sweep configuration.
0x14–0x20	ATK/DEC/SUS/REL	ADSR envelope parameters.
0x24	WAVE_TYPE	Waveform select (square, triangle, sine, saw, noise).
0x28	PWM_CTRL	PWM rate and depth.
0x2C	NOISEMODE	Noise generator algorithm (8 modes; see Chapter 12).
0x30	PHASE	Write to reset the phase.
0x34	RINGMOD	Ring-modulation source.
0x38	SYNC	Hard-sync source.
0x3C	DAC	Signed 8-bit sample (bypasses oscillator).

There are ten such channels, mapped at three base addresses:

Channels	Base	End
0–3	0xF0A80	0xF0B7F
4–6	0xF0C40	0xF0CFF
7–9	0xF0D40	0xF0DFF

Channels 0–3 are the primary block used by BASIC's **SOUND** keyword. Channels 4–9 are reachable directly from machine language; they form the "SID2" and "SID3" voice groups described in Chapter 15.

11.4 The SFX channels

In addition to the synth channels, there are four dedicated **SFX** channels that play raw samples from main memory. They live at 0xF0E80–0xF0EFF, with stride 0x20. Each channel:

Offset	Field	Meaning
0x00	SFX_PTR	Address of the sample data.
0x04	SFX_LEN	Sample length in bytes.
0x08	SFX_LOOP_PTR	Loop start (if looped).
0x0C	SFX_LOOP_LEN	Loop length.
0x10	SFX_FREQ	Playback rate in Hz.
0x14	SFX_VOL	Volume, 0–255.
0x18	SFX_FORMAT	0 = signed 8-bit, 1 = unsigned 8-bit, 2 = signed 16-bit.
0x1C	SFX_CTRL	Bit 0 = trigger, bit 1 = stop, bit 2 = loop enable.

A SFX_STATUS shadow register reads back bit 0 = playing, bit 1 = error.

11.5 The music-file player

Tracker and chip-tune music does not have to be poked into the audio chips one register at a time. Intuition Engine has a **media loader** that reads a file from disk, decodes it, and drives the appropriate audio engine on the program's behalf.

Address	Name	Purpose
0xF2300	MEDIA_NAME_PTR	Pointer to a null-terminated filename.
0xF2304	MEDIA_SUBSONG	Subsong number.
0xF2308	MEDIA_CTRL	1 = play, 2 = stop.
0xF230C	MEDIA_STATUS	0 = idle, 1 = loading, 2 = playing, 3 = error.

The BASIC keyword `SOUND PLAY "name"` uses this interface. The loader picks the right engine from the file extension. Chapter 22 lists every supported extension.

11.6 The global effects

Register	Address	Range	Purpose
FILTER_CUTOFF	0xF0820	0–255	Cutoff.
FILTER_RESONANCE	0xF0824	0–255	Q.
FILTER_TYPE	0xF0828	0–3	0 off, 1 LP, 2 HP, 3 BP.
FILTER_MOD_SOURCE	0xF082C	0–3	Modulation source channel.
FILTER_MOD_AMOUNT	0xF0830	0–255	Modulation depth.
OVERDRIVE_CTRL	0xF0A40	0–255	Drive amount, 0 = off.
REVERB_MIX	0xF0A50	0–255	Dry/wet.
REVERB_DECAY	0xF0A54	0–255	Tail length.

These effects sit at the end of the mixer chain and apply to the summed output of every engine.

11.7 BASIC keywords

Keyword	Routes to
<code>SOUNDch, freq, vol...</code>	SoundChip (Chapter 12)
<code>SOUND PLAY "file"</code>	Media loader (Chapter 22)
<code>SOUND STOP</code>	Media loader
<code>SOUND FILTER/REVERB/OVERDRIVE</code>	Mixer effects
<code>ENVELOPE</code>	SoundChip ADSR
<code>GATE</code>	SoundChip gate
<code>PSG...</code>	AY-3-8910 (Chapter 13)

Keyword	Routes to
SID...	SID family (Chapter 15)
POKEY...	POKEY (Chapter 17)
AHX...	AHX engine (Chapter 18)
SAP	POKEY SAP playback

11.8 What comes next

Chapter 12 covers the SoundChip and SFX in full. The remaining chapters of Part III take one chip each.